CSES — Tree Algorithms Reference

All tasks: 1 s, 512 MiB. Tree distances and path lengths are measured in edges.

1674 — Subordinates

Company hierarchy is rooted at employee 1. For every employee, count all descendants below them (not just direct reports). **In:** `n`; bosses of employees $2, \dots, n$

Out: n subordinate counts

Constraints: $n \le 2 \cdot 10^5$.

1130 — Tree Matching

Given an undirected tree, find the largest set of edges such that no vertex is incident to more than one chosen edge. **In:** `n`; then $n-1$ edges

Out: maximum matching size (number of edges)

Constraints: $n \le 2 \cdot 10^5$.

1131 — Tree Diameter

Find the maximum distance between any two vertices of an unweighted tree. **In:** `n`; then $n-1$ edges

Out: tree diameter in edges

Constraints: $n \le 2 \cdot 10^5$.

1132 — Tree Distances I

For every vertex v, find its eccentricity: the maximum distance from v to any other vertex. **In:** `n`; then $n-1$ edges

Out: n maximum distances, in vertex order

Constraints: $n \le 2 \cdot 10^5$.

1133 — Tree Distances II

For every vertex v, compute the sum of distances from v to all vertices of the tree. **In:** `n`; then $n-1$ edges

Out: n distance sums, in vertex order

Constraints: $n \le 2 \cdot 10^5$. Use 64-bit sums.

1687 — Company Queries I

The company hierarchy is rooted at employee 1. For each query (x, k), find the ancestor of employee x exactly k levels higher; report failure if it does not exist. **In:** `n q`; bosses of $2, \dots, n$; then queries `x k`

Out: the kth ancestor for each query, or `-1`

Constraints: $n, q \le 2 \cdot 10^5$, $1 \le k \le n$.

1688 — Company Queries II

For each pair of employees, find their lowest common ancestor in the hierarchy rooted at employee 1. **In:** `n q`; bosses of $2, \dots, n$; then queries `a b`

Out: LCA / lowest common boss for each query

Constraints: $n, q \le 2 \cdot 10^5$.

1135 — Distance Queries

For each query (a, b) on a fixed tree, report the number of edges on the unique path between a and b. **In:** `n q`; $n-1$ edges; then q pairs `a b`

Out: distance for each query

Constraints: $n, q \le 2 \cdot 10^5$.

1136 — Counting Paths

Given m queried tree paths, count for each vertex how many of those paths contain it. Both path endpoints count as contained vertices. **In:** `n m`; $n-1$ edges; then m endpoint pairs

Out: n path-containment counts

Constraints: $n, m \le 2 \cdot 10^5$.

1137 — Subtree Queries

Root the tree at 1; each vertex has a value. Support point assignment $v_s \leftarrow x$ and queries for the sum of values in the entire subtree of s (including s). **In:** `n q`; values; edges; queries `1 s x / 2 s`

Out: subtree sum for each type-2 query

Constraints: $n, q \le 2 \cdot 10^5$, values $\le 10^9$. Use 64-bit sums.

1138 — Path Queries

Root the tree at 1; each vertex has a value. Support point assignment $v_s \leftarrow x$ and queries for the sum on the inclusive path from root 1 to s. **In:** `n q`; values; edges; queries `1 s x / 2 s`

Out: root-to-s path sum for each type-2 query

Constraints: $n, q \le 2 \cdot 10^5$, values $\le 10^9$. Use 64-bit sums.

2134 — Path Queries II

Each tree vertex has a value. Support point assignment $v_s \leftarrow x$ and maximum-value queries on the inclusive path between arbitrary vertices a and b. **In:** `n q`; values; edges; queries `1 s x / 2 a b`

Out: path maximum for each type-2 query

Constraints: $n, q \le 2 \cdot 10^5$, values $\le 10^9$.

1139 — Distinct Colors

Root the tree at 1. For every vertex, count how many distinct color values occur among all vertices in its subtree, including itself. **In:** `n`; colors $c_1, \dots, c_n$; then $n-1$ edges

Out: n distinct-color counts

Constraints: $n \le 2 \cdot 10^5$, $c_i \le 10^9$.

2079 — Finding a Centroid

Find any vertex whose removal leaves every connected component with at most $\lfloor n/2 \rfloor$ vertices (equivalently, rooting there gives no child subtree larger than half the tree). **In:** `n`; then $n-1$ edges

Out: any centroid vertex

Constraints: $n \le 2 \cdot 10^5$.

2080 — Fixed-Length Paths I

Count distinct simple tree paths containing exactly k edges. A path is determined by its unordered pair of endpoints. **In:** `n k`; then $n-1$ edges

Out: number of length-k paths

Constraints: $1 \le k \le n \le 2 \cdot 10^5$. Use 64-bit count.

2081 — Fixed-Length Paths II

Count distinct simple tree paths whose length is between k_1 and k_2 edges, inclusive. A path is determined by its unordered pair of endpoints. **In:** `n k_1 k_2`; then $n-1$ edges

Out: number of paths with $k_1 \le$ length $\le k_2$

Constraints: $1 \le k_1 \le k_2 \le n \le 2 \cdot 10^5$. Use 64-bit count.